

[489] (continued)

On the Porism of the In-and-Circumscribed Polygon (continued).

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Now starting from the differential equation

$$\frac{dx}{\sqrt{(a, b, c, d, e \breve{a} x, 1)^4}} \pm \frac{dy}{\sqrt{(a, b, c, d, e \breve{a} y, 1)^4}} = 0,$$

the integral equation is known to be

$$\frac{\sqrt{(a, b, c, d, e \breve{a} x, 1)^4} - \sqrt{(a, b, c, d, e \breve{a} y, 1)^4}}{x - y} = \frac{1}{2}(a, b, c, d, e \breve{a} x, y, 1)^2,$$

where θ is the constant of integration. Writing, for shortness, $X = (a, b, c, d, e \breve{a} x, 1)^4$, $Y = (a, b, c, d, e \breve{a} y, 1)^4$, this is

$$X + Y - 2\sqrt{(XY)} = a(x^2 - y^2)^2 + 4b(x - y)(x^2 - y^2) + 6\theta(x - y)^2,$$

or, what is the same thing,

$$\begin{aligned} a(x + y)^4 - 2\sqrt{(XY)} &= a(x^2 - y^2)^2 - 4a(x - y)(x^3 - y^3) + 6\theta(x - y)^2, \\ &+ 4b(x^3 + y^3) \\ &+ 6c(x^2 + y^2) \\ &+ 4d(x + y) \\ &+ 2e, \end{aligned}$$

viz. this gives

$$\begin{aligned} \sqrt{(XY)} &= ax^2y^2 \\ &+ 2b(x^2y + xy^2) \\ &+ 3c(x^2 + y^2) \\ &+ 3d^2xy - 3\theta \\ &+ 2d(x + y) \\ &+ e, \end{aligned}$$

and, rationalising, the integral equation becomes

$$\begin{aligned} &= 6a\theta x^2y^2 \\ &- 4abxy(x + y) \\ &- ac(x + y)^2 \\ &+ 4b^2xy \\ &+ 12bcxy(x + y) + 12bdxy(x + y) \\ &- 9c^2xy \\ &+ 6ce(x + y)^2 - 18cd(x + y) \\ &- 12cd(x + y) \\ &+ 9d^2(x + y)^2 - 12d\theta(x + y) - 6ed + 4d^2 = 0. \end{aligned}$$

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or, as it may be written,

$$\begin{aligned}
 & xy^2(3\theta - 6ad) \\
 & + (x^2y + xy^2)(-4ad + 12bc - 12bd) \\
 & + (x^2 + y^2)(-ae - 3c^2 - 6ec\theta + 4d^2) \\
 & + xy(-2ae + 3bd - 12ec - 4be) \\
 & + (x + y)(-2be + 12bd - 6d^2) \\
 & + 4d^2 - 6ed = 0.
 \end{aligned}$$

Comparing this with the original integral equation $V = 0$, and the form of differential equation deduced therefrom, we ought to have therefore

$$\begin{aligned}
 & [(4b^2 - 8ad)x^2 + (-2ae + 4bd)x + (-ae + 4bd - 3c^2)]ay^2 \\
 & + [(-ae + 4bd - 4ce)x + (-2ae + 4bd - 4de) + (3d^2 - 6ed)]y \\
 & + [(-ae + 4bd - 3c^2)x + (3d^2 - 6ed) + (4e^2 - 8d\theta)]
 \end{aligned}$$

= multiple of X ,

= $(-4ax^2 - 4bx + 2cax)(\theta x + (3a - 6cx) - (-ae + 4bd - 3c^2))$, by comparing the coefficients of x^4 . I obtain this otherwise: Write

$$V = aC + 6\beta H,$$

then, forming the Hessian of V , we have

$$\square V = -aH^2 + (a^2I + (Ja\theta)U,$$

or

$$= aH^2 - \frac{3IJ(\theta)}{8}(\theta)(V - aC) + (aI + (Ja\theta)U,$$

that is,

$$aJV + (-3a^2J)V - \frac{2}{8}(aH - \frac{3J}{8})(aH) + \frac{3J^2(\theta)}{8}U(8H - aC) + (aJV + (Ja\theta)U,$$

viz.

$$K = -\frac{3J\theta}{8}(\theta) - aH,$$

this is

$$(aJV + (Ja\theta)V + (J^2I)V - (a\theta\theta)V + a\theta^2aV + \frac{3J}{8}(\theta) = (3aV)(-\theta U,$$

so that the components are

$$V = aV + 6\beta H,$$

$$V' = -aC + 6\beta H,$$

where $a, b, c, d, e \check{a} x, 1)^4, +(a, b, c, d, e \check{a} x, y) - 1)^2, 4ax + 4bx + 6cy - 4dy - e$, or $-d^2(c, 1)^2$,

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viz. the components are

$$\begin{array}{lll} (aa + b\check{a}(ax - bc), & ab + bd(ax - bc), & ae + bx + 2bd - 3c^2(\check{a}x, 1)^2, \\ (ab + bd(ax - bc), & ac + bx + 2bd - 3c^2(\check{a}x, 1)^2, & bd + bx + x_{1c}^2 J_x, \\ (ac + bx + 2bd - 3c^2(\check{a}x, 1)^2, & bd + bx + x_{1c}^2 J_x, & ae + bx + x_{1c}^2 J_x, \end{array}$$

where as before

$$K = \frac{8(a^2 - 3J\theta)}{a}.$$

I assume

$$\begin{aligned} B &= -\frac{1}{a}, \quad b' = bB - \frac{1}{a} \quad K = 3\{4a^2 - 6b\theta - 3a^2c^2\}, \\ ax + b\check{a}(ax - bc) &= a(x - bc) + ax(b - bx) - bx, \\ ab + bd(ax - bc) &= a(x - bc) + ax(b - bx) - bxc, \\ ae + bx + 2bd - 3c^2(\check{a}x, 1)^2 &= a(x - bc) + ax(b - bx) - bxc - bxd, \\ ae + bx + 2bd - 3c^2(\check{a}x, 1)^2 &= a(x - bc) + ax(b - bx) - bxc - bxd - bxe, \\ ax + b\check{a}(2bd - 3c^2)\check{a}J_x &= 4bK - 4bex + a(2bd - 3c^2)(\check{a}x, 1)^2 - \frac{1}{a}bK - 3bex, \\ &= -be - 12bc + 6b(8bd - 3c^2 + 4d\theta) \\ &= a(b\theta - 3b\theta + 3bax) - \frac{1}{2}(b\theta\theta + ac^2 + b\theta)f', \\ &= a(b\theta - 3b\theta + 3bax) - \frac{1}{2}(b\theta\theta + ac^2 + b\theta)f', \\ &= aa + b - 12acd + 6a^2ec, \end{aligned}$$

agreeing with the former result.

I return to the general form

$$\begin{aligned} &\eta(\theta', b, c \check{a}x, 1)^4 \\ &+ 2g(a', b', c' \check{a}x, 1)^2 \\ &+ (a'', b'', c'' \check{a}x, 1)^4 = 0 \end{aligned}$$

giving

$$\frac{dx}{\sqrt{(a, b, c, d, e \check{a}x, 1)^4}} \pm \frac{dy}{\sqrt{(a, b, c, d, e \check{a}y, 1)^4}} = \frac{dy}{\sqrt{(a', b', c' \check{a}y, 1)^2(a'', b'', c'' \check{a}y, 1)^2}}.$$

Operate a linear transformation on this, a say

$$x = \frac{\lambda x' + \mu}{\nu x' + \rho},$$

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the new coefficients are

$$\begin{aligned} & (a, b, c, d, e \check{a} x, 1)^4, \quad (a, b, c, d, e \check{a} x, 1)^4, \quad (a, b, c, d, e \check{a} x, 1)^4, \\ & (a', b', c' \check{a} x, 1)^2, \quad (a', b', c' \check{a} x, 1)^2, \quad (a', b', c' \check{a} x, 1)^2, \\ & (a'', b'', c'' \check{a} x, 1)^2, \quad (a'', b'', c'' \check{a} x, 1)^2, \quad (a'', b'', c'' \check{a} x, 1)^2; \end{aligned}$$

assume now

$$\begin{aligned} a'\lambda + (b' - c')\mu - ca &= 0, & ba\mu &= 0, \\ (a' + b')\lambda - bc + c\mu &= 0, & ca\mu &= a(a'\mu + 6c'\mu - bc) = 0, \\ b' & & & = c' = 6c\mu = 0; \end{aligned}$$

then it is easy to show that

$$\begin{aligned} (a, b, c \check{a} x, p\mu \check{a} x, 1)^4 &= (a', c' \check{a} x, 1)^2, \\ (a', b', c' \check{a} x, p\mu \check{a} x, 1)^2 &= (a', c' \check{a} x, 1)^2, \\ (a, b, c \check{a} x, p\mu \check{a} x, 1)^4 &= (a', c' \check{a} x, 1)^2, \end{aligned}$$

and the equations give

$$\begin{vmatrix} a' & b' - c' & \mu & a \\ b' + c' & b' - c' & \mu & ba \\ a' + b' & a' - b' & ca\mu & a \\ b'' & c'' & a'' - c'' & 1 \end{vmatrix} = 0;$$

that is

$$\begin{aligned} & (a'c'' - b''^2 + a''b'^2)(a'c''(c' - b')) - 2b'^2 \\ & \quad - (c'^2c'' - b'c''c''c'' - b'^2) \\ & \quad + a'c''(a''a' - a''b'c' - a'b'c) \\ & \quad + a'c''(a'c''^2c''c'' - b'^2c'') \\ & \quad + a'c''(a'b'^2 + a''b'') \\ & = 0, \end{aligned}$$

which is

$$\begin{aligned} (c'c'' - b''^2)(a'c'' - b''^2) &= \frac{1}{4}b'^2 & a, \quad b, & c \\ & + ca''c''^2 & a', \quad b', & c' \\ + ca'c''^2 - 2(c''^2 - c'') & & a'', \quad b'', & c'' \\ & + c''(c''^2 - c'') \\ & + 2bc(b - c) \\ + 2bc(b - c)(c'' - c''^2) & & & \\ + 4b'c''(b'c'^2 - c'^2) & & & \\ + 8c'(b'^2 - c''^2) + d\theta & = 0. \end{aligned}$$

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If the original matrix be symmetrical, that is

$$(\beta - b')^2 + (a\mu - b')(c\mu - f'') = \begin{vmatrix} a & b & f \\ h & b & f \\ p & f & c \end{vmatrix}, \quad \text{this is}$$

$$\begin{aligned} c(\beta - b')^2 + (a\mu - b')(c\mu - f'') &= -\partial F \begin{vmatrix} a & b & g \\ h & b & f \\ p & f & c \end{vmatrix} \\ &+ h^2(-c\eta) \\ &+ b^2(-c\mu - g^2 - 2fh) \\ &+ f^2(-c\nu) \\ &+ 2h(c^2g + ph) \\ &+ 2f^2(ch + ck) \\ &+ 2hc(fp + ch) + 6h = 0, \end{aligned}$$

that is

$$\begin{aligned} (b - g)(b - b')^2 + (a + b'')(c\beta + p) + 2(cp^2 + ch^2 - 2bfh) \\ = 2F(abc - aef^2 - bg^2 - ch^2 + 2fgh) + b'(4h^2 - bf^2 + bh) - cf'' + b^2 = 0, \end{aligned}$$

multiplied by

$$\theta + b - g = 0,$$

viz. the equation in θ is

$$(\theta + b - g)(b'\theta^2 + (b - c)\theta + (4fh - bcb + hc(2g - b) - c)) + (b'^2 - cab) + 2(cb^2 + ch^2 - 2hf)\beta = 0.$$

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